



Report

Early age at menopause and breast cancer: are leaner women more protected? A prospective analysis of the Dutch DOM cohort

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Summary

To investigate the relationship between age at menopause, body mass index, and breast cancer risk, we used data from a prospective cohort study (DOM cohort) in the Netherlands. Participants in this breast cancer-screening project included 10,591 women living in Utrecht, aged 49–66 years at enrolment. During a median follow-up period of 19 years, women attended screening rounds at which anthropometric measurements were taken and questions were asked about menopausal status, age at menopause, medication use and other risk factors for breast cancer. Cox regression analysis was used to investigate the relationship between age at menopause and subsequent breast cancer risk. Breast cancer incidence decreased with an earlier age at menopause. Women with a menopausal age of 44 years or younger had a 34% lower risk of breast cancer, than women with a menopausal age over 54 years (hazard ratio is 0.66 (95% confidence interval 0.43–0.91)). The annual hazard of breast cancer incidence decreased by 2.6% per year reduction in age at menopause. The protective effect of an early age at menopause was stronger for women with a low body mass index (≤ 27 kg/m²; reduction of 44%) than for women with a high body mass index (>27 kg/m²; reduction of 24%), although this difference was not statistically significant (P for interaction = 0.58). This difference was most pronounced in women who had ever smoked. Adjustment for known breast cancer risk factors did not alter the crude risk estimates significantly.

In conclusion, this study provides evidence of the protective effect of lower age at menopause on subsequent breast cancer risk. This protective effect may be even stronger in leaner women.

Introduction

The incidence of breast cancer among women in the Netherlands is the highest in Europe. One of every 10 women will develop breast cancer sometime during her life and one of every 22 women will die of the disease [1]. Since breast cancer has a large impact on society, many epidemiological studies have been performed to explain the etiology of the disease. Factors related to endogenous hormones, such as menarche and menopause, are frequently investigated as risk factors for breast cancer and many studies have indicated the protective effect of early menopause. The underlying hypothesis is that a longer exposure to high levels of endogenous estrogens and possibly progesterones is related to an increased risk of breast cancer [2, 3]. These hormones are supposed to act at an

early stage in the development of breast cancer by promoting the excessive proliferation of normal epithelial stem cells [4].

Even though the ovaries cease the production of estrogens after menopause, serum estrogen levels can remain impressive, principally due to the conversion of androstenedione and testosterone to estrogen in fat cells [5]. In fact, postmenopausal overweight women generally have higher levels of biologically active estrogens than postmenopausal women of normal weight [6]. Because of this, obesity is considered a risk factor for breast cancer in postmenopausal women [7–10]. One may, therefore, hypothesize that an early age at menopause is especially protective in lean women, since they may have a more dramatic decline in endogenous hormone production than fatter women. One case-control study addressing this hypothesis showed

a stronger protective effect of early menopause in leaner women [11], whereas two other studies did not [8, 12].

The purpose of the present study was to investigate the relationship between age at menopause, body mass index (BMI) and breast cancer risk in a well-defined cohort of postmenopausal women. This relationship was studied in a prospective cohort design, based on data collected in the 1970s and 1980s in a population-based breast cancer-screening program in the Netherlands.

Patients and methods

A population-based breast cancer-screening project (the DOM project) was started in December 1974 in Utrecht, the Netherlands [13]. The women born between 1911 and 1925 were 49–66 years of age at recruitment. Baseline measurements included a mammography, a physical examination and the collection of information through self-administered questionnaires concerning health, reproductive history, and family history. A breast cancer-screening registry collected data on the occurrence of breast cancer in these DOM participants. All breast cancer cases in this study were histologically confirmed primary cancers. The local authority informed the investigators on a monthly basis about the deaths and movements of cohort members.

A total of 14,697 women participated in the DOM project (72% of the invited women). Data from 4106 of these women were excluded because (1) they had not reached menopause at baseline or during the study ($n = 1583$), (2) premenopausal breast cancer developed ($n = 2$), (3) they had breast cancer at baseline ($n = 63$), or (4) data on smoking were missing ($n = 2458$) (information about smoking habits was not asked for at the beginning of the project). Therefore, data from 10,591 women were available for analysis.

Age at menopause was defined as having had no menstrual period for at least 12 consecutive months. Women who had had a hysterectomy or surgical removal of the ovaries were coded as having an artificial menopause. Age at menopause was analyzed continuously and categorically (1) ≤ 44 , (2) 45–49, (3) 50–54, and (4) ≥ 55 years and was reported at first screening visit or calculated from information during follow-up visits.

Anthropometric measurements, taken while the women were standing without shoes and upper clothes, included height, weight, and subscapular and triceps skinfold measurements. Body height was measured to the nearest 0.5 cm and weight to the nearest 0.5 kg. BMI was computed as weight (kg) divided by height squared (m^2). Since skinfold thicknesses were measured with callipers that could not measure skinfolds thicker than 40 mm, these skinfolds were coded as 40 mm on the registration forms. The study population was divided into subgroups by quintiles of subscapular and triceps skinfolds [14], resulting in subgroups of women with equal fat distribution, peripheral fat distribution, and central fat distribution. Women with central fat distribution were defined as women with fat predominantly located on the trunk with high subscapular skinfold thickness and relatively low triceps skinfold thickness. Women with peripheral fat distribution had high triceps skinfold thickness and relatively low subscapular skinfold thickness [14].

Socio-economic status (SES) was based on the type of medical health insurance: sick fund (lower status), civil servant (intermediate), or private (higher status). Parity was defined as the number of live births.

Analyses were adjusted for age at study entry (continuous), height (continuous), fat distribution (central, other), parity (0, 1–2, 3–4, ≥ 5), SES (low, intermediate, high), smoking habits (ever, never), use of oral contraceptive pills (ever, never), and age at first birth (no children, ≤ 24 , > 24). Follow-up was performed until January 1, 1996. The end points noted during the study included incidence of primary breast cancer, death, and lost to follow-up because of moving to other communities.

We assessed the effect of menopausal age (in 4 strata) on subsequent breast cancer risk using the Cox proportional hazards analysis. Age at menopause of 55 years or older was the reference category. The simultaneous effect of age at menopause and the potential confounders was investigated to see whether the crude hazard ratio of age at menopause changed substantially. A test for linear trend in hazard ratios over the menopausal age categories was also conducted.

The effect of menopausal age was investigated continuously. The presence of an interaction between BMI and age at menopause was investigated (1) by comparing the hazard ratios for the menopause classes in the two strata of BMI ($\leq 27 \text{ kg/m}^2$; $> 27 \text{ kg/m}^2$) and (2) by comparing models with and without the product term (BMI * age at menopause).

Table 1. Characteristics of the study population according to menopausal age

Class		Age at menopause				Total (<i>n</i> total)
		≤ 44	45–49	50–54	≥ 55	
		<i>n</i> = 1626 mean (SD)	<i>n</i> = 3363 mean (SD)	<i>n</i> = 4848 mean (SD)	<i>n</i> = 754 mean (SD)	
Age at entry		56.7 (4.4)	56.8 (4.4)	57.6 (3.9)	58.0 (4.2)	10,591
Height		162.3 (6.3)	162.3 (6.1)	162.0(6.1)	161.9(6.2)	10,587
Parity	No children	20.0%	18.6%	17.4%	16.4%	1916
	1–2 children	38.9%	34.7%	34.0%	31.7%	3685
	3–4 children	27.0%	30.8%	31.0%	33.3%	3232
	≥ 5 children	14.1%	15.9%	17.6%	18.6%	1758
						(10,591)
Health insurance	Sick fund	69.7%	68.5%	65.4%	63.1%	7082
	Civil servant insurance	12.1%	11.5%	12.2%	13.0%	1271
	Private insurance	18.2%	20.0%	22.4%	23.9%	2236
						(10,589)
Smoking	Never	67.8%	68.6%	74.1%	80.2%	7377
	Yes/ever	32.2%	31.4%	25.9%	19.8%	2891
						(10,268)
Oral contraceptives	Never	97.4%	93.4%	92.5%	91.9%	9903
	Yes/ever	2.6%	6.6%	7.5%	8.1%	668
						(10,591)
Age at first birth	No children	20.0%	18.6%	17.4%	16.4%	1916
	≤ 24 years	32.8%	30.7%	28.1%	27.2%	3131
	> 24 years	47.2%	50.7%	54.5%	56.4%	5539
						(10,586)
BMI	≤ 27	68.6%	71.2%	67.8%	65.0%	7281
	> 27	31.4%	28.8%	32.2%	35.0%	3305
						(10,586)
Fat distribution	Central fatty distribution	11.1%	11.8%	12.9%	14.5%	1311
	Other	88.9%	88.2%	87.1%	85.5%	9255
						(10,566)
Type of menopause	Natural	47.4%	75.1%	90.0%	91.4%	8343
	Artificial	52.6%	24.9%	10.0%	8.6%	2247
						(10,590)

The effect of age at menopause was also investigated in subgroups: women with a natural menopause versus an artificial menopause and ever-smokers versus never-smokers. The subgroups of ever-smokers and never-smokers were further stratified by BMI to assess whether the effects within these groups differed by BMI.

All statistical analyses were performed using SPSS for Windows software.

Results

Table 1 shows the baseline characteristics of the study population divided into four menopausal age categories. The mean age of the DOM participants at entry

was 57 years (standard deviation (SD) 4.3), mean BMI was 26.2 kg/m² (SD 3.8), and mean age at menopause was 48.8 years (SD 4.6). Approximately 21.2% of the women had an artificial menopause. As expected, crude data showed a decreasing proportion of artificial menopause with increasing age at menopause. Ever-users of oral contraceptives were rare among these women, who were aged 37–52 when oral contraceptives were introduced in the Netherlands.

Table 2 shows the follow-up results of the 10,591 women in the cohort. A total of 485 (4.6%) women developed breast cancer, 2046 (19.3%) died, and 1558 (14.7%) moved between the date of enrolment and January 1, 1996 (177,523 person years of observation). The proportion of women who had moved

Table 2. Follow-up of 10,591 women divided into menopausal age categories

	Total	Age at menopause			
		≤ 44	45–49	50–54	≥ 55
Persons at risk	10,591	1626	3363	4848	754
Person-years at risk	177523.2	27232.6	56450.3	81267.1	12573.2
Total died (%)	2046 (19.3)	312 (19.2)	662 (19.7)	931 (19.2)	141 (18.7)
Moved (%)	1558 (14.7)	255 (15.7)	507 (15.1)	687 (14.2)	109 (14.5)
Breast cancer incidence (%)	485 (4.6)	62 (3.8)	139 (4.1)	238 (4.9)	46 (6.1)

Table 3. Hazard ratio of age at menopause (continuous) and breast cancer incidence adjusted for potential confounders

Crude	Hazard ratio (95% CI)
Adjusted for all potential confounders	0.974 (0.954–0.994)
Adjusted for single potential confounders	
Age at entry	0.977 (0.957–0.997)
Height	0.974 (0.954–0.994)
Parity	0.974 (0.954–0.994)
Health insurance	0.974 (0.954–0.994)
Smoking	0.978 (0.957–0.998)
Oral contraceptives	0.974 (0.954–0.994)
Age at first birth	0.975 (0.955–0.995)
BMI	0.975 (0.955–0.995)
Fat distribution	0.975 (0.954–0.994)
Type of menopause	0.975 (0.953–0.996)

was distributed equally among the menopausal age categories.

The breast cancer incidence decreased with an earlier age at menopause (Table 3). We quantified the continuous hazard ratio of an earlier age at menopause to be 0.974 (95% confidence interval (CI) 0.954–0.994) (Table 3), which means that each year when menopause is earlier decreases the annual hazard of breast cancer incidence by 2.6%.

Individually, none of the potential confounders influenced the estimation of the hazard ratio of menopausal age (Table 3). When the hazard ratio was adjusted for all potential confounders simultaneously (0.977 (95% CI 0.956–0.999)), confounding was also negligible. Therefore, we will further report crude risk estimates.

As a single risk factor in the Cox regression model, BMI was associated with an increased risk of

breast cancer in this cohort. Women with a high BMI ($> 27 \text{ kg/m}^2$) had a risk of 1.34 (95% CI 1.11–1.61) of developing postmenopausal breast cancer compared to women with a lower BMI (data not shown).

Table 4 shows the estimates of the hazard ratios of age at menopause for all women in the cohort, for women with a low BMI ($\leq 27 \text{ kg/m}^2$), and with a high BMI ($> 27 \text{ kg/m}^2$). There was an obvious decreasing trend in hazard ratios with earlier age at menopause in the total population. Women who were relatively young at menopause (≤ 44) had a statistically significant lower breast cancer risk (0.66 (95% CI 0.43–0.91) than women who were older at menopause (≥ 55). Furthermore, the preventive effect of an earlier menopausal age was stronger in leaner women (BMI $\geq 27 \text{ kg/m}^2$) than in fatter women (BMI $> 27 \text{ kg/m}^2$). Leaner women with menopause at age of 44 or younger had a decreased breast cancer risk of 44% compared to women with menopause at age 55 or older. This decrease was 24% for fatter women with an early menopause. The test for linear trend was significant for leaner women ($P = 0.013$), but not for fatter women ($P = 0.28$). However, the differences in the hazard ratios of the BMI strata were not statistically significant ($P = 0.58$).

Previous studies have shown that smoking causes menopause to occur earlier. In the DOM cohort, menopause of smoking women occurred, on average, 8 months earlier. We also found that the difference in the effect of menopausal age on breast cancer risk was especially pronounced in smoking women (risk reduction of 21% in heavy smoking women and 67% in leaner smoking women) (Table 5).

The protective effect on breast cancer risk due to earlier menopause is somewhat stronger (but not statistically significant) for women with an artificial menopause than for women with a natural menopause (data not shown).

Table 4. Relation between age at menopause and breast cancer incidence for the total population and for subgroups of BMI¹

Age at menopause	Hazard ratio (95% CI)		
	Total population <i>n</i> = 10, 591	BMI ≤ 27 <i>n</i> = 7281	BMI > 27 <i>n</i> = 3305
55	1	1	1
50–54	0.80 (0.58–1.10)	0.75 (0.50–1.11)	0.91 (0.54–1.52)
45–49	0.67 (0.48–0.94)	0.65 (0.43–0.99)	0.74 (0.42–1.29)
≤ 44	0.66 (0.43–0.91)	0.56 (0.34–0.91)	0.76 (0.41–1.40)
Test for trend	<i>P</i> = 0.0072	<i>P</i> = 0.0129	<i>P</i> = 0.2808

¹Crude hazard ratios are presented.

Table 5. Age at menopause and breast cancer risk according to combined smoking status and BMI

Age at menopause	Hazard ratio (95% CI)						
	Total population	Never smoked	Ever smoked	Never smoked	Never smoked	Ever smoked	Ever smoked
	<i>n</i> = 10, 591	<i>n</i> = 7, 377	<i>n</i> = 2, 891	BMI ≤ 27 <i>n</i> = 4, 945	BMI > 27 <i>n</i> = 2, 429	BMI ≤ 27 <i>n</i> = 2, 138	BMI > 27 <i>n</i> = 735
≤ 44	0.66 (0.43–0.91)	0.72 (0.46–1.12)	0.45 (0.21–0.96)	0.70 (0.39–1.24)	0.76 (0.38–1.52)	0.33 (0.13–0.83)	0.79 (0.20–3.17)
45–49	0.67 (0.48–0.94)	0.76 (0.52–1.12)	0.50 (0.25–0.97)	0.74 (0.45–1.22)	0.79 (0.43–1.46)	0.42 (0.19–0.92)	0.64 (0.18–2.34)
50–54	0.80 (0.58–1.10)	0.87 (0.60–1.25)	0.57 (0.30–1.10)	0.89 (0.56–1.42)	0.84 (0.47–1.49)	0.41 (0.19–0.88)	1.05 (0.31–3.50)
≥ 55	1	1	1	1	1	1	1

Discussion

Our results show that an earlier age at menopause has a protective effect on the risk of breast cancer. The strongest and most significant protective effects were observed when menopause occurred at age 45–49 or at age 44 or younger (hazard ratios of 0.67 and 0.66, respectively) than when it occurred at age 55 or older. The results are consistent with the hypothesis of a causal relationship with hormonal changes occurring at this point in life. The cumulative life time exposure of these women to endogenous hormones is shorter than in women with a late menopause.

The MEDLINE bibliographic database from January 1966 to January 1998 was used to identify English language publications assessing the effect of menopause on subsequent breast cancer risk. A total of 14 case-control and 2 cohort studies that assessed this relationship showed the protective effect of early menopause [11, 12, 15–17, 20–30]. We found a 2.6% decrease in breast cancer risk per year reduction in age at menopause. This result is comparable with two prospective studies showing decreases of 3.5% and 3.0%, respectively [15, 16], and a case-control study

showing a 4.0% decrease per year reduction in age at menopause [17].

The present study suggests that leaner women benefit more from earlier menopause than fatter women, although the interaction term is not statistically significant. A similar, but statistically significant, indication was noted in a case-control study [11]. This was the only study that specifically assessed the modifying effect of BMI on the relationship between age at menopause and breast cancer. Two other studies reported that a significant interaction between BMI and menopausal age on the incidence of breast cancer could not be found [8, 12]. However, no risk estimates for different BMI strata were given in these two studies, so we were not able to see whether the effects were in the same direction as in our study. A possible explanation for the modifying effect of BMI might be the more dramatic decline in endogenous hormone production in leaner women than in fatter women. Besides the ovarian production of endogenous hormones, obese women may have an impressive (continuous) production in their adipose tissue.

In the DOM cohort, smoking lowered the age of menopause by approximately 8 months. Therefore,

with the above results in mind, it can be hypothesized that if smoking women have lower estrogen levels [31] and lean smoking women have the most dramatic decline in these hormones after menopause, then early age at menopause is particularly protective in leaner women who have ever smoked, as was found in our study.

The women were asked at enrolment about use of exogenous hormones during the preceding 12 months. Only 10.2% of the women answered this question positively. Exclusion of these women from the analysis did not substantially alter the estimates of the hazard ratios.

At study entry, 85% of the women were already postmenopausal. It is possible that these women misclassified their menopausal age. However, since menopause is considered to be an important event by most women, it is usually correctly remembered [18, 19]. Furthermore, in this prospective study, potential misclassification was nondifferential and may have resulted in an underestimation of the risks. Analyses limited to women with recent menopausal events did not substantially alter the result, suggesting no major attenuation effect of misclassification.

The Cox regression model adjusts for censored data. It is based on the assumption that the outcome risk of withdrawn women does not differ from that of women in the cohort. If withdrawal is associated with breast cancer and/or menopausal age, it may have resulted in selective drop-out. We do not expect this, because withdrawal did not differ substantially between the menopausal age categories (approximately 15% in each menopausal stratum). In addition, the mean menopausal age of the withdrawn women was 48.7 years, almost identical to that of the women in the cohort (48.8 years).

In conclusion, this prospective study provides evidence of the protective effect of early age at menopause on subsequent breast cancer risk, and suggests a stronger effect for leaner women than for fatter women. This effect was most pronounced among smoking women.

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